

DRAWING INDEX

DEVICE SYMBOL LEGEND

WIRING STANDARDS

REVID	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
B	CD	POWER CORD NOTES FOR NVIDIA PCB	09/13/24

DRAWING NUMBER	DESCRIPTION
1	TITLE PAGE
1A	POWER PLUGS
1B	SYSTEM NETWORK TOPOLOGY
2	POWER DISTRIBUTION
3	OPT022 IO #1
4	OPT022 IO #2
5	NETWORK TOPOLOGY
6	PANEL LAYOUT

SYMBOL	DESCRIPTION
BC	BULKHEAD CONNECTOR
C	CONTACTOR
CB	CIRCUIT BREAKER
CR	CONTROL RELAY
CCR	CLEANROON ROBOT
DISC	DISCONNECT SWITCH
DS	DOOR SWITCH
E	ELECTRICAL ENCLOSURE
EPB	EMERGENCY STOP PUSHBUTTON
ER	EMERGENCY STOP RELAY
EF	EXHAUST FAN
FB	FUSE BLOCK
FU	FUSE
FWC	FIELD WIREABLE CONNECTOR
GS	GUARD SWITCH
H	SIGNAL HORN
HMI	HUMAN TO MACHINE INTERFACE
IO	INPUT/OUTPUT MODULE
IM	PLC INPUT MODULE
JB	JUNCTION BOX
KS	KEY SWITCH
MET	METER
RC	ROBOT CONTROLLER
LT	LIGHT
M	MOTOR
MS	MOTOR STARTER
MSF	MOTOR STARTER FORWARD
MSR	MOTOR STARTER REVERSE
MC	MOTOR CONTROLLER
OM	PLC OUTPUT MODULE
POT	POTENTIOMETER
PB	PUSHBUTTON
PC	PERSONAL COMPUTER
PDB	POWER DISTRIBUTION BLOCK
PE	PHOTO EYE
PLC	PROGRAMMABLE LOGIC CONTROLLER
PRX	PROXIMITY SWITCH
PS	PRESSURE SWITCH OR POWER SUPPLY
REC	RECEPTACLE
RC	ROBOT CONTROLLER
SR	SAFETY RELAY
SS	SELECTOR SWITCH
SC	SPLITTER CABLE
SV	SOLENOID VALVE
SW	OPERATOR SWITCH
SM	SAFETY MODULE
T	TRANSFORMER
TB	TERMINAL BLOCK
TL	TOOL
TS-X	TERMINAL STRIP("X" DENOTES NUMBER)
UPS	UNINTERRUPTABLE POWER SUPPLY
VFD	VARIABLE FREQUENCY DRIVE
VC	VISION CONTROLLER
HATM	HINE ATMOSPHERIC ROBOT
DV	DEVICE

DESCRIPTION	WIRE COLOR
POWER	BLACK
AC CONTROL CIRCUITS, 120VAC	RED
NEUTRAL, GROUNDED CIRCUIT CONDUCTOR	WHITE
24VDC CONTROL CIRCUIT	BLUE
24VDC CONTROL CIRCUIT COMMON	BLUE/WHITE
CIRCUIT SUPPLIED FROM EXTERNAL POWER SOURCE	YELLOW
EQUIPMENT GROUNDING CONDUCTOR – INSULATED	GREEN/YEL
MULTI CONDUCTOR CABLE	blk, wht, etc.

DESCRIPTION	MINIMUM GAUGE
POWER CIRCUITS	14
LIGHTING AND CONTROL CIRCUITS ON MACHINE	16
CONTROL CIRCUITS WITHIN ENCLOSURE	18
ELECTRONIC PROGRAMMABLE CONTROL I/O IN WIREWAY	24
ELECTRONIC PROGRAMMABLE CONTROL I/O WITHIN ENCL	26
POWER EARTH GROUND – I.E., CHASSIS GROUND, ETC.	18

IMPORTANT NOTES

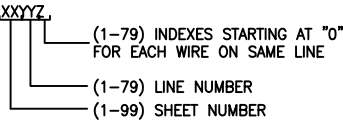
SIGNAL WIRING DUCTS SHALL CONTAIN LOW VOLTAGE, DC CONTROL WIRING (24VDC) AND BELOW. POWER WIRING DUCTS SHALL CONTAIN ALL POWER AND CONTROL WIRING 120VAC AND ABOVE.

ALL WIRES AND CABLES SHALL BE LABELED AT EACH END WITH WIRE NUMBER ASSIGNED AS PER ELEMENTARY DIAGRAMS.

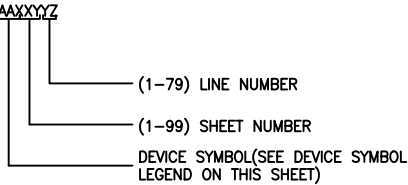
WIRE LABELS SHALL HAVE COMPLETE WIRE NUMBER DISPLAYED ON SINGLE LINE. WIRE NUMBER TO BE REPEATED ON THREE OR MORE LINES.

ALL PANEL DEVICES SHALL BE LABELED WITH ADHESIVE LABEL SHOWING DEVICE CODE

WIRE LABEL DESIGNATOR



DEVICE LABEL DESIGNATOR



STANDARD WIRE SIZE	SIZE	AWG or kcmil		COPPER	14	12	10	8	6	4	3	2	1	1/0	2/0	3/0	4/0	250	300	350	400	500
	75 C (167 F)				RH, RHW, THHW, THW, THWN, XHHW, ZW																	
	90 C (194 F)	THHN, THHW, THW-2, HWN-2, RHH, RWH-2, USE-2, XHHW, XHHW-2, ZW-2																				



1040 SERPENTINE LANE, SUITE 208 PLEASANTON, CA ,94566 (925) 344-1221	DESIGNER CD	CREATED 09/03/24	DRAWING TITLE TRADE SHOW STANDS TITLE PAGE	CUSTOMER RED HAT	JOB NO. 10013
CONFIDENTIAL INFORMATION OF KANAVU AUTOMATION LLC NOT TO BE USED, COPIED OR DISCLOSED TO OTHERS WITHOUT WRITTEN CONSENT.	ENGINEER CD	CHECKED		DRAWING NO. STAND 1	SHEET 1 OF 6

REVID	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
B	CD	POWER CORD NOTES FOR NVIDIA PCB	09/13/24

Stand 1 - Europe
Type F Power Strip provided by Customer
CORD, POWER, TYPE F, 8FT LNG Cut off the C13 end and tie to power supply
POWER SUPPLY, 5A, 24VDC
SWITCH, ETHERNET, UNMANAGED, 5 PORTS
BOX, PUSHBUTTON, 5 HOLES
CONTROLLER, OPT22,
MOTOR, STEPPER
STACKLIGHT, 24VDC, COLUMN MOUNT
DC TO DC CONVERTER, 24VDC TO 10VDC
POTENTIOMETER, 5KOHM, 22MM, BLACK
CAMERA, 48MP, DIGITAL, MICROSCOPE, 4K. Includes Barrel Connector to US Plug. ADDING TYPE F ADAPTER PLUG
CAMERA LIGHT comes with US Plug. ADDING TYPE F ADAPTER PLUG
JESTON ORIN NANO DEVELOPER KIT: Comes with 19VDC Power Supply with Type F Plug

Stand 2 - Europe
Type F Power Strip provided by Customer
1. Advantech Provided power adapter with F type plugs
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
2. Advantech Provided power adapter with F type plugs
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
3. Advantech Provided power adapter with F type plugs
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
4. 24V POWER ADAPTER SET with C13 to CEE7/7 Power Cordset
CONNECTOR, BARREL TO SCREW
SWITCH, ETHERNET, UNMANAGED, 5 PORTS

Stand 3 - Europe
Type F Power Strip provided by Customer
1. Advantech Provided power adapter with F type plugs
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
2. 24V POWER ADAPTER SET with C13 to CEE7/7 Power Cordset
CONNECTOR, BARREL TO SCREW
SWITCH, ETHERNET, UNMANAGED, 5 PORTS
3. 48V POWER ADAPTER SET with C13 to CEE7/7 Power Cordset
CONNECTOR, BARREL TO SCREW
FORTINET ETHERNET SWITCH
4. Dell Provided C13 to Type F Power Cordset
Dell Server

Stand 1 - Asia
Type G Power Strip provided by Customer
CORD, POWER, TYPE G, 6FT Cut off the C13 end and tie to power supply
POWER SUPPLY, 5A, 24VDC
SWITCH, ETHERNET, UNMANAGED, 5 PORTS
BOX, PUSHBUTTON, 5 HOLES
CONTROLLER, OPT22,
MOTOR, STEPPER
STACKLIGHT, 24VDC, COLUMN MOUNT
DC TO DC CONVERTER, 24VDC TO 10VDC
POTENTIOMETER, 5KOHM, 22MM, BLACK
CAMERA, 48MP, DIGITAL, MICROSCOPE, 4K. Includes Barrel Connector to US Plug. ADDING TYPE G ADAPTER PLUG
CAMERA LIGHT comes with US Plug. ADDING TYPE G ADAPTER PLUG
JESTON ORIN NANO DEVELOPER KIT: Comes with 19VDC Power Supply with Type G Plug
ADAPTER, US TO Type G
32' TV

Stand 2 - Asia
Type G Power Strip provided by Customer
1. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
2. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
3. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
4. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
SWITCH, ETHERNET, UNMANAGED, 5 PORTS
ADAPTER, US TO Type G
32' TV

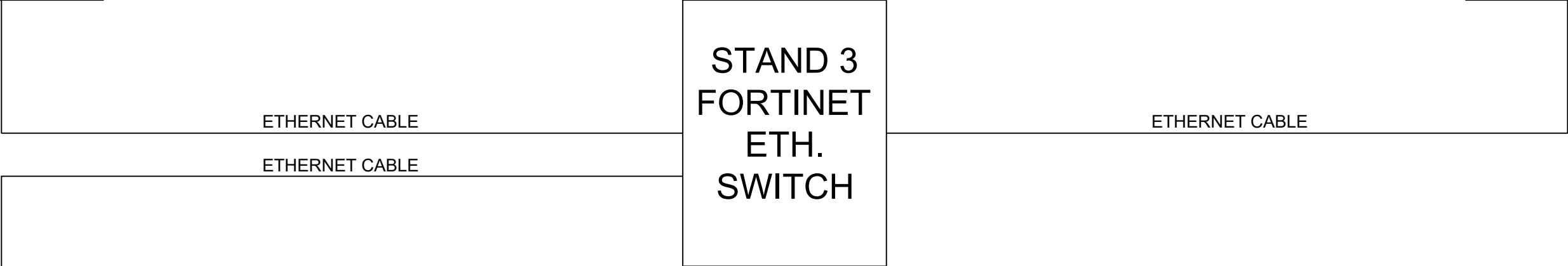
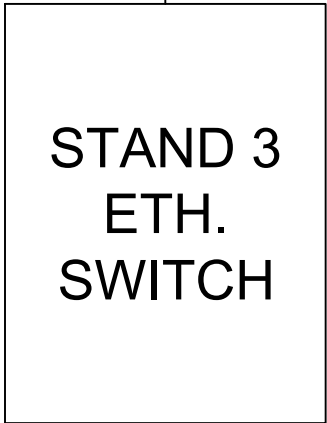
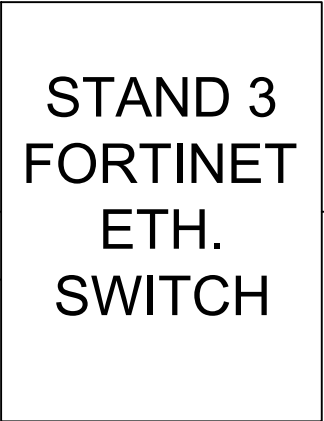
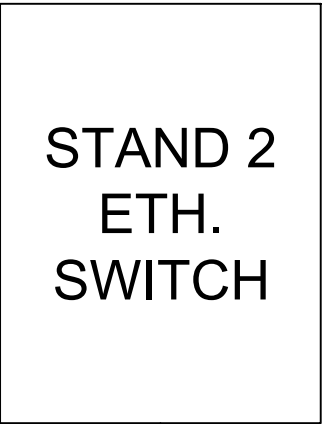
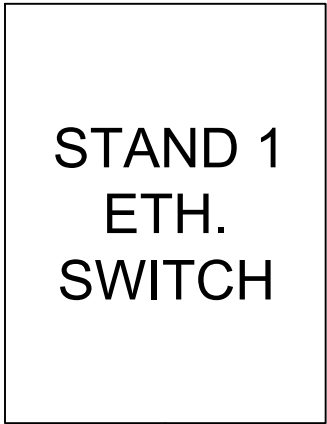
Stand 3 - Asia
Type G Power Strip provided by Customer
1. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
ADVANTECH PC
2. 24V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
SWITCH, ETHERNET, UNMANAGED, 5 PORTS
3. 48V POWER ADAPTER SET with C13 to Type G Power Cordset
CONNECTOR, BARREL TO SCREW
FORTINET ETHERNET SWITCH
4. Dell Provided C13 to Type G Power Cordset
Dell Server
ADAPTER, US TO Type G
32' TV

STANDARD WIRE SIZE	SIZE	AWG or kernl		COPPER	14	12	10	8	6	4	3	2	1	1/0	2/0	3/0	4/0	250	300	350	400	500
	75 C (167 F)				RH, RHW, THHW, THW, THWN, XHHW, ZW	15	20	30	50	65	85	100	115	130	150	175	200	230	255	285	310	335
	90 C (194 F)	THHN, THHW, THW-2, HWN-2, RHH, RWH-2, USE-2, XHHW, XHHW-2, ZW-2	15		20	30	55	75	95	115	130	145	170	195	225	260	290	320	350	380	430	
	TYPES																					



1040 SERPENTINE LANE, SUITE 208 PLEASANTON, CA ,94566 (925) 344-1221	DESIGNER CD	CREATED 09/03/24	DRAWING TITLE TRADE SHOW STANDS	CUSTOMER RED HAT	JOB NO. 10013
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REVID	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
B	CD	POWER CORD NOTES FOR NVIDIA PCB	09/13/24

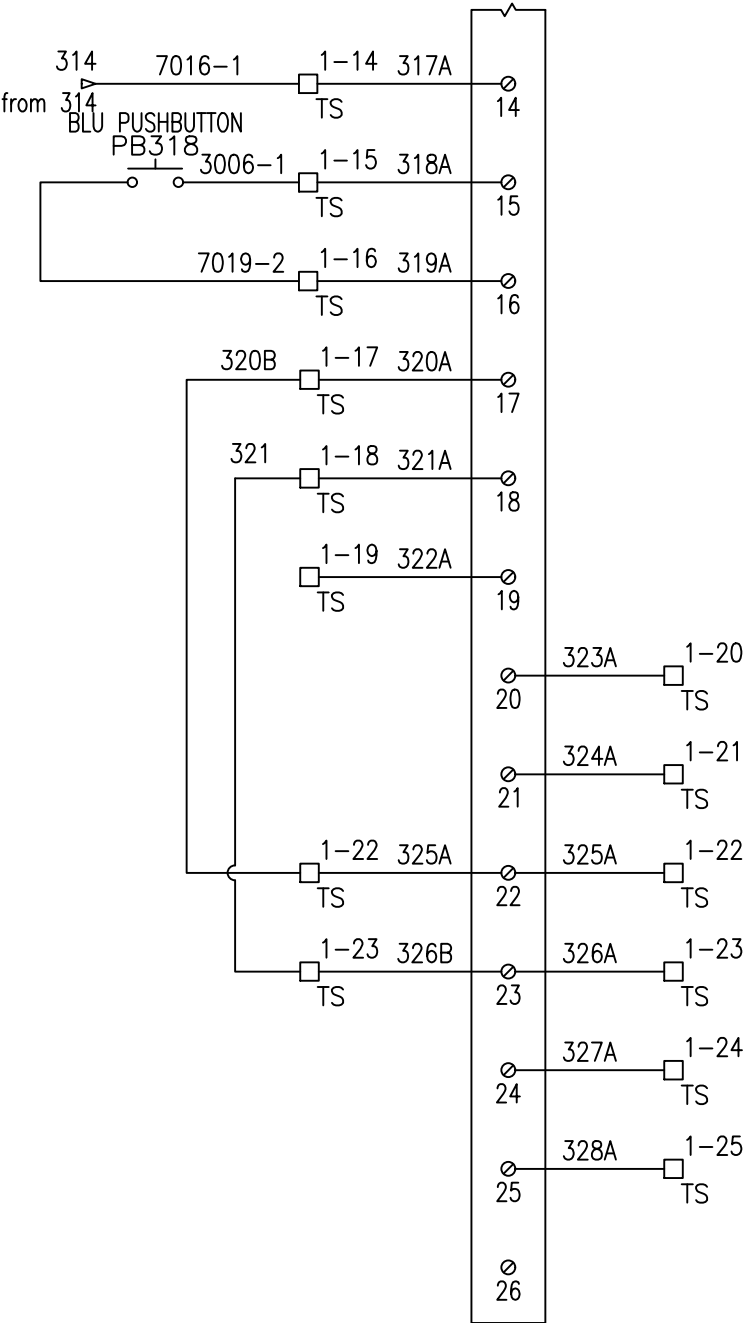
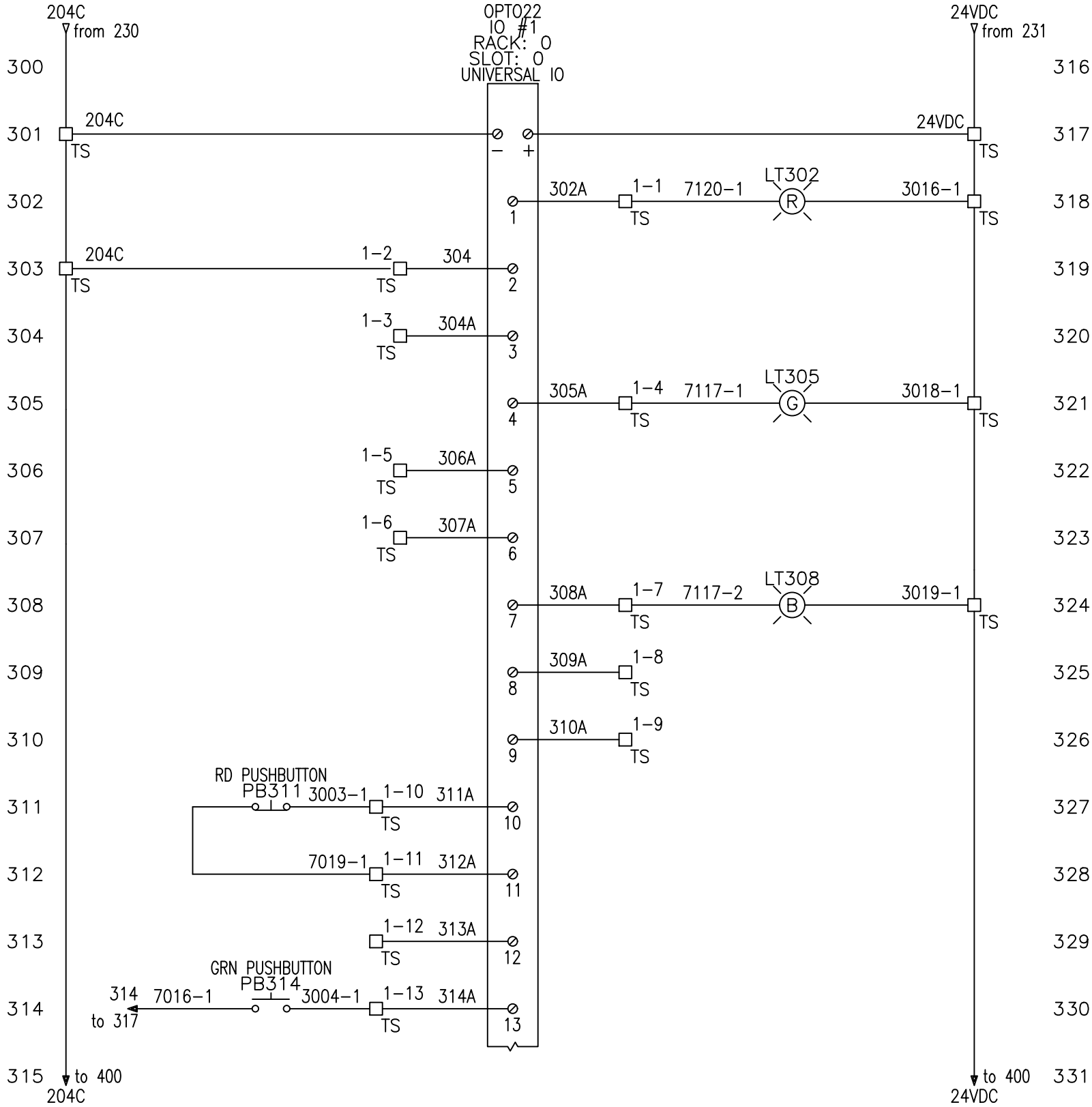


STANDARD WIRE SIZE	SIZE	AWG or Kcmil		COPPER	14	12	10	8	6	4	3	2	1	1/0	2/0	3/0	4/0	250	300	350	400	500
	75 C (167 F)	TYPES	RH, RHW, THHW, THW, THWN, XHHW, ZW		15	20	30	50	65	85	100	115	130	150	175	200	230	255	285	310	335	380
	90 C (194 f)		THHN, THHW, THW-2, HWN-2, RHH, RWH-2, USE-2, XHHW, XHHW-2, ZW-2		15	20	30	55	75	95	115	130	145	170	195	225	260	290	320	350	380	430



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REVID	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
B	CD	POWER CORD NOTES FOR NVIDIA PCB	09/13/24
C	CD	REDLINES	10/07/24



STANDARD WIRE SIZE	SIZE	AWG or kcmil																COPPER	14	12	10	8	6	4	3	2	1	1/0	2/0	3/0	4/0	250	300	350	400	500	
	75 C (167 F)	TYPES	RH, RHW, THHW, THW, THWN, XHHW, ZW																15	20	30	50	65	85	100	115	130	150	175	200	230	255	285	310	335	380	
			THHN, THHW, THW-2, HWN-2, RHH, RWH-2, USE-2, XHHW, XHHW-2, ZW-2																15	20	30	55	75	95	115	130	145	170	195	225	260	290	320	350	380	430	
			90 C (194 F)																																		



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1040 SERPENTINE LANE, SUITE 208
PLEASANTON, CA 94566
(925) 344-1221

DESIGNER
CD

ENGINEER
CD

CREATED
09/03/24

CHECKED

DRAWING TITLE
TRADE SHOW STANDS
OPTO22 IO #1

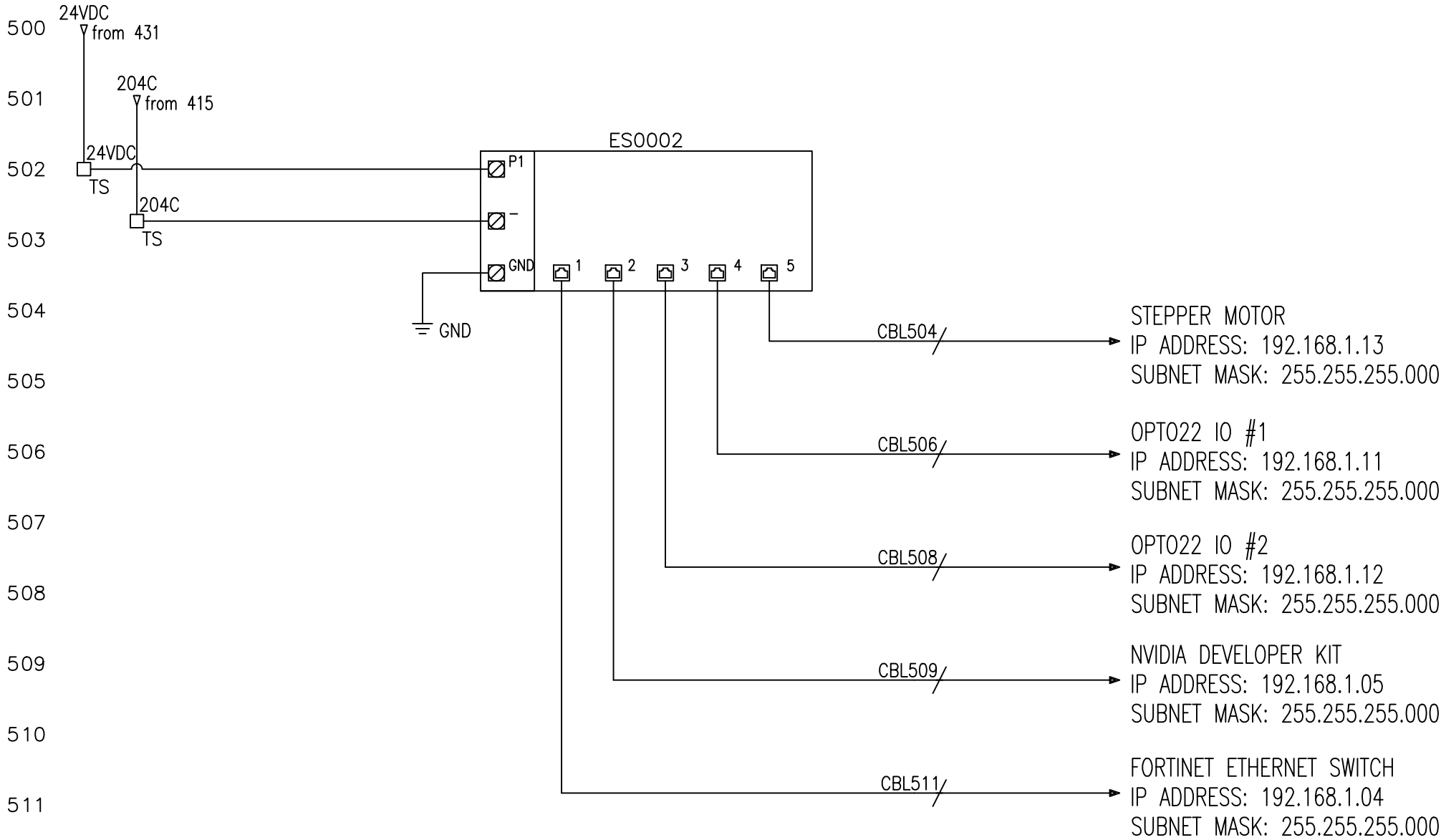
CUSTOMER
RED HAT

DRAWING NO.
STAND 1

JOB NO.
10013

SHEET
3 OF 6

REV/D	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
B	CD	POWER CORD NOTES FOR NVIDIA PCB	09/13/24
C	CD	REDLINES	10/07/24



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				15	20	30	50	65	85	100	115	130	150	175	200	230	255	285	310	335	380
				15	20	30	55	75	95	115	130	145	170	195	225	260	290	320	350	380	430



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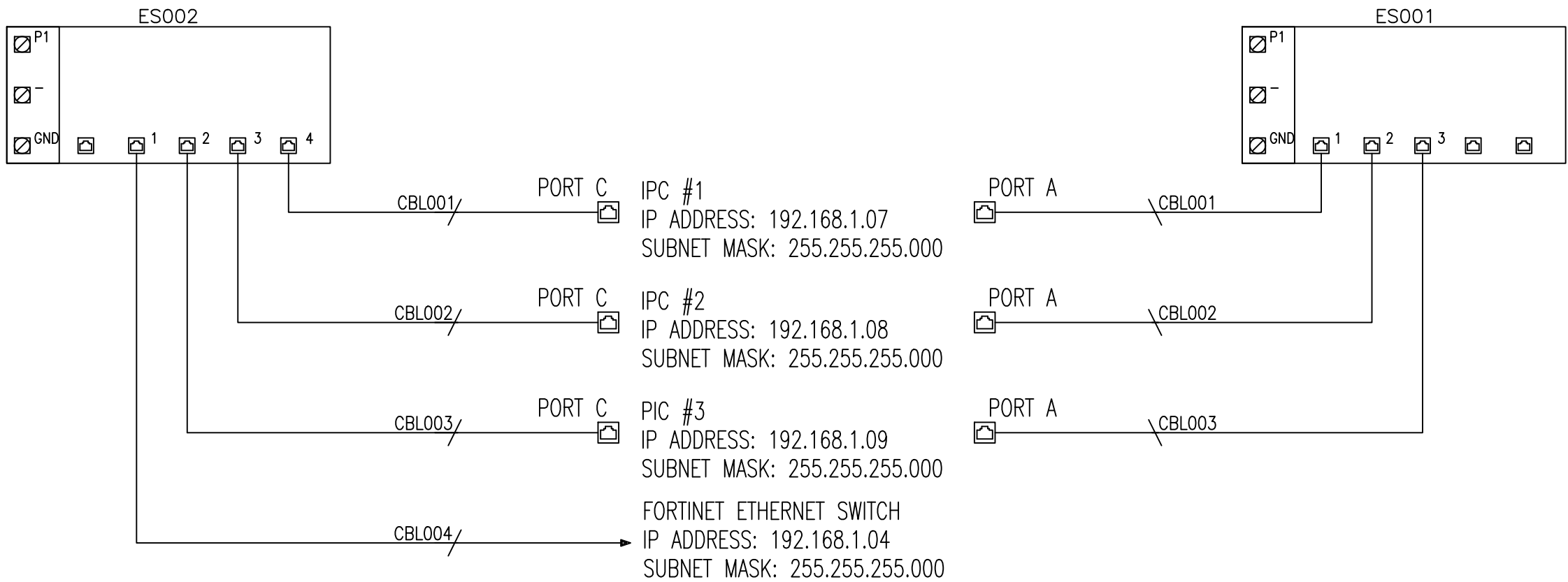
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NOTES:

POWER TO EACH DEVICE WILL COME FROM AN AC TO DC
ADAPTER PLUGGED INTO CUSTOMER PROVIDED POWER STRIP.

WIRE WILL NEED TO BE LANDED FROM BARREL ADAPTERS TO
POWER TERMINALS OF THE IPCS. COLOR CODE SHOULD FOLLOW
STANDARDS LOCATED ON TITLE PAGE OF THIS DRAWING.

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A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
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C	CD	REDLINES	10/07/24



STANDARD WIRE SIZE	SIZE	AWG or kernl	COPPER	14	12	10	8	6	4	3	2	1	1/0	2/0	3/0	4/0	250	300	350	400	500	
	TYPES			RH, RHW, THHW, THW, THWN, XHHW, ZW																		
				THHN, THHW, THW-2, HWN-2, RHH, RWH-2, USE-2, XHHW, XHHW-2, ZW-2																		
	75 C (167 F)			15	20	30	50	65	85	100	115	130	150	175	200	230	255	285	310	335	380	
	90 C (194 F)			15	20	30	55	75	95	115	130	145	170	195	225	260	290	320	350	380	430	



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1040 SERPENTINE LANE, SUITE 208 PLEASANTON, CA ,94566 (925) 344-1221	DESIGNER CD	CREATED 09/03/24	DRAWING TITLE TRADE SHOW STANDS NETWORK TOPOLOGY	CUSTOMER RED HAT	JOB NO. 10013
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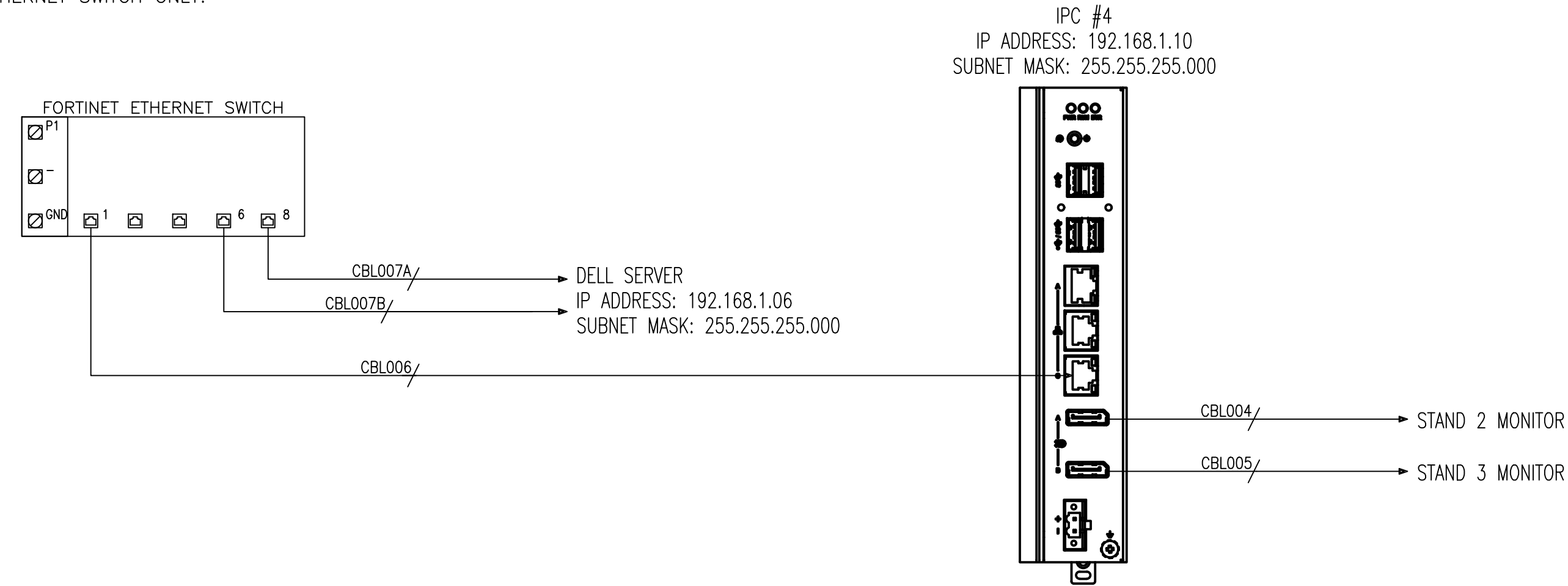
NOTES:

POWER TO EACH DEVICE WILL COME FROM AN AC TO DC
ADAPTER PLUGGED INTO CUSTOMER PROVIDED POWER STRIP.

WIRE WILL NEED TO BE LANDED FROM BARREL ADAPTERS TO
POWER TERMINALS OF THE IPCS. COLOR CODE SHOULD FOLLOW
STANDARDS LOCATED ON TITLE PAGE OF THIS DRAWING.

48VDC POWER ADAPTER SHOULD BE INSTALLED TO POWER THE
FORTINET ETHERNET SWITCH ONLY.

REVID	BY	DESCRIPTION	DATE
A	CD	ADDED: NETWORK PAGE & POWER CORD	09/05/24
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